

Loading graphPT and namelistPT

Cleaning the names

Getting all redirects

Computing strong forward links (can be ignored for loading)

```
In[*]:= Clear[WikiPageIDs, WikiTitles, StrongForwardLinks]
WikiPageIDs[arts_List, lang_String] := Module[{},
  Inner[Rule, arts, WikipediaData[arts, "PageID", Language → lang], List]] /;
  Length[arts] ≤ 30;
WikiPageIDs[arts_List, lang_String] := Module[{},
  Flatten[Join @@ Map[WikiPageIDs[#, lang] &, Partition[arts, UpTo[30]]]]] /;
  Length[arts] > 30;
WikiTitles[pageIDs_List, lang_String] := Module[{goodIDs, badIDs},
  goodIDs = Union[Select[pageIDs, IntegerQ]];
  badIDs = Complement[pageIDs, goodIDs];
  Union[Join[Inner[Rule, goodIDs, WikipediaData["PageID" → goodIDs, "Title",
    Language → lang], List], Map[#, Missing["NotAvailable"] &, badIDs]]]
] /; Length[pageIDs] ≤ 30;
WikiTitles[pageIDs_List, lang_String] := Module[{},
  Flatten[Join @@ Map[WikiTitles[#, lang] &, Partition[pageIDs, UpTo[30]]]]] /;
  Length[pageIDs] > 30;
StrongForwardLinks[v_, lang_String] := Module[{fv, wikiIDs, fvPageIDs,
  redirectPageIDs, redirects, redtitles, text, test, strongconnections},
  Print["Strong forward links for " <> v];
  Pause[1.5];
  fv = EdgeList[graphPT2, v → _] [[All, 2]];
  If[fv == {}, Return[{}]];
  wikiIDs = WikiPageIDs[fv, lang];
  If[wikiIDs == {}, Return[{}]];
  fvPageIDs = Select[wikiIDs, IntegerQ[#[[2]]] &];
  redirects = GetRedirects[fvPageIDs[[All, 2]], lang];
  redirectPageIDs = Map[Rule[#[[1]], Append[#[[2]], #[[1]]] &, redirects];
  redtitles = Map[#[[1, 2]] → #[[2, All, 2]] &, redirectPageIDs];
  text = WikipediaData[v, "ArticleWikicode", Language → lang];
  test[l_List] :=
    StringContainsQ[text, Map["[" <> # <> "]" | "[" <> # <> "|" ~ ~ __ ~ ~ "]" &, l]];
  strongconnections = Select[redtitles, test[#[[2]]] &] [[All, 1]];
  Map[v → # &, Union[strongconnections]]
]

In[*]:= Clear[level0, StrongForwardLinks1]
level0 = namelistPT;
StrongForwardLinks1 = Map[StrongForwardLinks[#, "Portuguese"] &, level0];
Put[level0, path <> "level0.dat"]
Put[StrongForwardLinks1, path <> "StrongForwardLinks1.dat"]
```

```

In[ ]:= Clear[StrongGraphForwardLevel1, OriginalGraphForwardLevel1]
StrongGraphForwardLevel1 =
  Graph[Join @@ Map[EdgeList[Graph[Join @@ StrongForwardLinks1], # &→ _] &, level0],
    ImageSize → 800];
Put[StrongGraphForwardLevel1, path <> "StrongGraphForwardLevel1.dat"]
OriginalGraphForwardLevel1 =
  Graph[Join @@ Map[EdgeList[graphPT, # &→ _] &, level0], ImageSize → 800];
Put[OriginalGraphForwardLevel1, path <> "OriginalGraphForwardLevel1.dat"]

```

Computing the strong linked graph

```

In[ ]:= Clear[graphPT2]
graphPT2 = Get[path <> "graphPT2.dat"]

```

Out[]:=

Graph[ Vertex count: 33 961
Edge count: 252 882]

```

In[ ]:= Clear[v1]
v1 = Union[EdgeList[graphPT2][[All, 1]]]
Length[v1]

```

Out[]:=

{Aage Niels Bohr, Aarão Burlamaqui Benchimol, Aarne Haapakoski, Aaron Copland, Aaron Moten, Aaron Ramsey, Aaron Sorkin, Aava Santiago, Abade Faria, Abas Mirza, Abbé Pierre, Abbie Hoffman, Abby Mann, Abd al-Ilah, Abdas de Susa, Abdellah Taïa, Abdelmajid Lakhali, Abdelwahab Meddeb, ... 29 221 ..., Zoroastro Gouveia, Zózimo Bulbul, Zózimo Tavares Mendes, Zuca Sardan, Zuenir Ventura, Zulaiê Cobra, Zuleika Alambert, Zulmira Maria de Jesus, Zulmira Miranda, Zulmira Ribeiro Tavares, Zumbi dos Palmares, Zuria Vega, Zuza Homem de Mello, Zuzu Angel, Zuzuca do Salgueiro, Zuzu Cavaquinho, Zwi Skornicki, Zygmunt Bauman}

Full expression not available (original memory size: 1.6 MB)



Out[]:=

29 257

```

In[ ]:= Clear[partv1, k]
partv1 = Partition[v1, UpTo[50]];
k = Length[partv1]

```

Out[]:=

586

```

In[ ]:= Clear[sf1]
Do[Pause[0.9];
  sf1[i] = Map[StrongForwardLinks[#, "Portuguese"] &, partv1[[i]]];
  Put[sf1[i], path <> "sf1-" <> ToString[i] <> ".dat"];
  Print["Batch " <> ToString[i] <> " done!";, {i, 1, k}]

In[ ]:= sf1[231] = Map[StrongForwardLinks[#, "Portuguese"] &, partv1[[231]]];
Put[sf1[231], path <> "sf1-" <> ToString[231] <> ".dat"];

In[ ]:= Do[Pause[0.9];
  sf1[i] = Map[StrongForwardLinks[#, "Portuguese"] &, partv1[[i]]];
  Put[sf1[i], path <> "sf1-" <> ToString[i] <> ".dat"];
  Print["Batch " <> ToString[i] <> " done!";, {i, 253, k}]

In[ ]:= sf1[356] = Map[StrongForwardLinks[#, "Portuguese"] &, partv1[[356]]];
Put[sf1[356], path <> "sf1-" <> ToString[356] <> ".dat"];

```

```

In[ ]:= Do[Pause[0.9];
  sfl[i] = Map[StrongForwardLinks[#, "Portuguese"] &, partv1[[i]]];
  Put[sfl[i], path <> "sfl-" <> ToString[i] <> ".dat"];
  Print["Batch " <> ToString[i] <> " done!"];, {i, 383, k}]

In[ ]:= Do[Pause[0.9];
  sfl[i] = Map[StrongForwardLinks[#, "Portuguese"] &, partv1[[i]]];
  Put[sfl[i], path <> "sfl-" <> ToString[i] <> ".dat"];
  Print["Batch " <> ToString[i] <> " done!"];, {i, 409, k}]

In[ ]:= sfl[419] = Map[StrongForwardLinks[#, "Portuguese"] &, partv1[[419]]];
Put[sfl[419], path <> "sfl-" <> ToString[419] <> ".dat"];

In[ ]:= Do[Pause[0.9];
  sfl[i] = Map[StrongForwardLinks[#, "Portuguese"] &, partv1[[i]]];
  Put[sfl[i], path <> "sfl-" <> ToString[i] <> ".dat"];
  Print["Batch " <> ToString[i] <> " done!"];, {i, 433, k}]

In[ ]:= sfl[535] = Map[StrongForwardLinks[#, "Portuguese"] &, partv1[[535]]];
Put[sfl[535], path <> "sfl-" <> ToString[535] <> ".dat"];

```

Constructing the strong linked graph

```

Clear[sfl]
Do[sfl[i] = Get[path <> "sfl-" <> ToString[535] <> ".dat"], {i, 1, k}];

```

```

In[ ]:= Clear[strongedges]
strongedges = Union[Flatten[Join@@Table[sfl[i], {i, 1, k}]]]

```

Out[]:=

```

{Aarão Burlamaqui Benchimol ↔ Miguel Couto, Aaron Copland ↔ André Gide, Aaron Copland ↔ Edward Weston,
 Aaron Copland ↔ Elliot Goldenthal, Aaron Copland ↔ Jean-Paul Sartre, Aaron Copland ↔ Marcel Proust,
 Aaron Copland ↔ Pablo Picasso, Aaron Copland ↔ Paul Valéry, ... 55459 ..., Zuza Homem de Mello ↔ João Gilberto,
 Zuzu Angel ↔ Chico Buarque, Zuzu Angel ↔ José Louzeiro, Zuzu Angel ↔ Zezé Polessa,
 Zuzu Cavaquinho ↔ João do Rio, Zwi Skornicki ↔ Salvador Dalí, Zygmunt Bauman ↔ Theodor W. Adorno}

```

Full expression not available (original memory size: 8.5 MB)



```

In[ ]:= Clear[graphPT3]
graphPT3 = Graph[strongedges];
Put[graphPT3, path <> "graphPT3.dat"]

Clear[graphPT3]
graphPT3 = Get[path <> "graphPT3.dat"]

```

```
In[*]:= NeighborhoodGraph[graphPT3, "Carolina Maria de Jesus", VertexLabels -> "Name"]
Out[*]=
```

